

Mark schemes

Q1.

(a) $1 - e^{-\frac{1}{\mu}x} = \frac{1}{4}, \quad \frac{3}{4}$

Either.

M1

$$Q_1 = \mu \ln \left(\frac{4}{3} \right)$$

M1 for attempting either Q_1 or Q_3 .

M1A1

$$Q_3 = \mu \ln 4$$

A1

$$\text{IQR} = \mu \ln 3$$

CAO

A1

5

(b) $E(X^2) = \int_0^{\infty} \frac{1}{\mu} x^2 e^{-\frac{1}{\mu}x} dx$

Knowledge of formula.

M1

$$= \left[-x^2 e^{-\frac{1}{\mu}x} \right]_0^{\infty} + \int_0^{\infty} 2x e^{-\frac{1}{\mu}x} dx$$

Using integration by parts.

M1A1

$$= 0 + 2\mu \cdot \mu = 2\mu^2 \quad (\text{AG})$$

A1

4

(c) $\text{SD} = \sqrt{2\mu^2 - \mu^2} = \mu$

B1

$$\ln 3 > 1 \Rightarrow \text{SD} < \text{IQR}$$

M1A1

3

[12]

Q2.

(a) $E(X^2) = \int_0^{\infty} kx^2 e^{-kx} dx$

M1

$$= \left[-x^2 e^{-kx} \right]_0^{\infty} + \int_0^{\infty} 2xe^{-kx} dx$$

M1

$$= 0 + \left[-\frac{2x}{k} e^{-kx} \right]_0^{\infty} + \int_0^{\infty} \frac{2}{k} e^{-kx} dx$$

0 may be omitted, or limits inserted at end of process.

(E(X) integral can be quoted.)

A1

$$= 0 + \left[-\frac{2}{k^2} e^{-kx} \right]_0^{\infty}$$

Ditto.

A1

$$= \frac{2}{k^2}$$

A1

$$\text{Var}(X) = \frac{2}{k^2} - \left(\frac{1}{k} \right)^2 = \frac{1}{k^2}$$

*Their $E(X^2)$ minus mean², provided **positive**.*

SC Allow B1 for those who write correct working and result, having failed to integrate correctly

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A1 ✓

6

(b) (i) $F(x) = \int_0^x k e^{-ku} du$

M1

$$\left[-e^{-ku} \right]_0^x = 1 - e^{-kx}$$

A1A1

3

(ii) $\left[1 - e^{-kx} \right]_0^N = 0.9 \Rightarrow e^{-kN} = 0.1$

M1

$$\Rightarrow N = \frac{1}{k} \ln 10$$

M1A1

3

(c) Mean = $a = \frac{1}{k} \Rightarrow k = \frac{1}{a},$

M1A1

$$\text{Mean} = 3a = \frac{1}{k} \Rightarrow k = \frac{1}{3a}$$

$$e^{-\frac{1}{a}a} \cdot e^{-\frac{1}{3a}a} = e^{-1} \cdot e^{-\frac{1}{3}} = e^{-\frac{4}{3}}$$

CWO

M1A1

4

[16]

Q3.

(a) (i) $\mu = E(X) = \int_0^{\infty} kxe^{-kx} dx$

M1

$$= \left[-xe^{-kx} \right]_0^{\infty} + \int_0^{\infty} e^{-kx} dx$$

M1

$$= 0 + \left[\frac{e^{-kx}}{-k} \right]_0^{\infty}$$

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A1

$$= \frac{1}{k}$$

AG

3

(ii) $\int_0^m ke^{-kx} dx = \frac{1}{2}$

Or $F(x) = 1 - e^{-kx}$

M1

$$\left[-e^{-kx} \right]_0^m = \frac{1}{2}$$

$$-e^{-km} + 1 = \frac{1}{2}$$

$$\frac{1}{2} = 1 - e^{-km}$$

A1

$$e^{-km} = \frac{1}{2} \Rightarrow km = \ln 2 \Rightarrow m = \frac{\ln 2}{k}$$

M1A1

$$\ln 2 < 1 \Rightarrow m < \mu$$

A1

5

(b) (i) $P(T = 0) = e^{-\frac{t}{\lambda}}$

B1

1

(ii) (A) $P(T > t) = 1 - F(t) = e^{-\frac{t}{\lambda}}$

M1

$$P(T < t) = F(t) = 1 - e^{-\frac{t}{\lambda}} \quad t \geq 0$$

AG

A1

2

(B) $f(t) = F'(t) = \frac{1}{\lambda} e^{-\frac{t}{\lambda}} \quad t \geq 0$

M1

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⇒ Exponential distribution

A1

2

[13]

Q4.

(a) $F(x) = 1 - e^{-\lambda x}, x \geq 0$

$$F(x) = 1 - e^{-\lambda x} \quad \text{B1B0}$$

B1

$$F(x) = 0, x < 0$$

Dependent

B1

2

(b) $1 - e^{-\lambda x} = \frac{3}{4}$

For either Q_1 or Q_3

M1

$$Q_3 = \frac{1}{\lambda} \ln 4$$

m1 for attempting to solve for either Q_1 or Q_3

m1A1

$$1 - e^{-\lambda x} = \frac{1}{4}$$

$$Q_1 = \frac{1}{\lambda} \ln \frac{4}{3}$$

$$\text{IQR} = \frac{1}{\lambda} \ln 3$$

AG

A1

A1

5

(c) (i) $E(X^2) = \int_0^{\infty} \lambda x^2 e^{-\lambda x} dx$

Limits required

M1

$$= \left[-x^2 e^{-\lambda x} \right]_0^{\infty} + \int_0^{\infty} 2x e^{-\lambda x} dx$$

A1

$$= \left[-\frac{2x}{\lambda} e^{-\lambda x} \right]_0^{\infty} + \int_0^{\infty} \frac{2}{\lambda} e^{-\lambda x} dx$$

A1

$$= \left[-\frac{2}{\lambda^2} e^{-\lambda x} \right]_0^{\infty}$$

A1

4

$$= \frac{2}{\lambda^2}$$

AG

$$(ii) \quad \text{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

AG

B1

1

$$(d) \quad (i) \quad \frac{1}{\lambda} \ln 3 = \frac{4}{\lambda^2}$$

M1

$$\lambda = \frac{4}{\ln 3}$$

A1

2

$$(ii) \quad \text{IQR} \rightarrow 0 \text{ as } \lambda \rightarrow \infty$$

E1

1

[15]

Q5.

$$(a) \quad E(X) = \int_0^{\infty} \lambda x e^{-\lambda x} dx$$

Use of

M1

$$= \left[-x e^{-\lambda x} \right]_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx$$

Integrate by parts

A1

$$= \left[\frac{-e^{-\lambda x}}{\lambda} \right]_0^{\infty}$$

Correctly

A1

$$= \frac{1}{\lambda}$$

AG

A1

4

(b) (i) $\frac{1}{\alpha} = \frac{1}{62.5} = 0.016$

M1A1

2

(ii) $F(t) = 1 - e^{-0.016 \times 80}$
 Or $\int_{80}^{\infty} 0.016 e^{-0.016t} dt$

Award M1A1 for complement

M1A1

$P(T > 80) = e^{-0.016 \times 80}$

M1

$= 0.278$

A1

4

(iii) **Either**
 $e^{-0.016 \times 100} + e^{-0.016 \times 80}$

M1A1

$= 0.726$

A1

or
 $e^{-0.016 \times 20}$

(M2)

$= 0.726$

(A1)

3

[13]

Q6.

(a) (i) $f(t) = \frac{1}{800} e^{-\frac{t}{800}}$
 CAO

B1

$f(t) = 0$ for $t \geq 0$
 for $t < 0$

CAO both

B1

- (ii) $\sigma = 800$
CAO

B1

1

(iii) $P(T > +1000) = \int_{1000}^{\infty} \frac{1}{800} e^{-\frac{t}{800}} dt$
Use of $\int f(t) dt$

M1

$$= \left[-e^{-\frac{t}{800}} \right]_{1000}^{\infty}$$

integral & limits

A1

$$= e^{-\frac{1000}{800}} = 0.286 \text{ to } 0.287$$

AWFW

A1

3

- (iv) $P(T > +1000 | T > 1000) =$
Use of $P(A \& B) / P(A)$

M1

$$\frac{P(T > 2000)}{P(T > 1000)} =$$

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OE

A1

$$\frac{\left[-e^{-\frac{t}{800}} \right]_{2000}^{\infty}}{\left[-e^{-\frac{t}{800}} \right]_{1000}^{\infty}} = 0.286 \text{ to } 0.287$$

AWRT

If answer (iv) = answer (iii), then 3 marks
if answer (iii) correct, otherwise 2 marks

A1

3

- (b) Number of components working after 1000 hours ~ binomial

M1

with $n = 4$ and $p = 0.286/7$
f.t. on (iii)

A1f.t.

$P(X \geq 2) = P(X = 2, 3, 4) =$
use of either

M1

$1 - P(X = 0, 1) =$

0.323 to 0.326

AWFW

A1

4

[13]

Q7.

(a) $SD(T) = 8$

cao

B1

1

(b) $P(5 < T < 15) =$
 $P(T < 15) - P(T < 5)$

Use of; or use of $\int \lambda e^{-\lambda t} dt$

M1

$$= \left(1 - e^{-\frac{15}{8}}\right) - \left(1 - e^{-\frac{5}{8}}\right)$$

$$\text{Or } \left[-e^{-\lambda t}\right]_5^{15}$$

≥ 1 correct term

A1

$$= e^{-\frac{5}{8}} - e^{-\frac{15}{8}}$$

$$= 0.53526 - 0.15335 = 0.382$$

awrt

A1

3

(c) $P(\text{none from 5.45 to 6.00}) = P(T > 15)$
Attempt at either probability

M1

$$= e^{-\frac{15}{8}} = 0.153$$

f.t. on part (b) (0.153355)

A1f.t.

$$P(\text{none from 6.00 to 6.15}) = P(S > 15)$$

$$= e^{-1} = 0.368$$

cao/awrt (0.367879)

A1

$$P(\text{none from 5.45 to 6.15}) = \text{'product'}$$

Use of

m1

$$= e^{-\frac{23}{8}} = 0.056 \text{ to } 0.057$$

awfw

A1

5

Or (using Poisson) =

(M1)

$$P(0 \mid \lambda = \frac{15}{8} = 1.875)$$

(A1)

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$$P(0 \mid \lambda = \frac{15}{8} = 1.875)$$

(A1)

$$= e^{-1.875} \times e^{-1}$$

Product

(m1)

$$= e^{-2.875} \times 0.056 \text{ to } 0.057$$

awfw

(A1)

[9]

Examiner reports

Q1.

Students were far happier with this question, where their pure mathematics ability enabled them to deal well with logarithms and the integration that was required to answer this question. Almost all seemed familiar with the statistical knowledge required. It was noticed however that 'range' was sometimes used when 'interquartile range' was meant. Since both are well-defined concepts, with distinctly different meanings, this could not be accepted. Otherwise, the few marks that were lost resulted from small matters of detail, for example sign errors, or misuse of brackets. It was pleasing to note that the vast majority of students realised that $3 > 1$ was needed to answer the last part of the question correctly.

Q2.

Most students were able to express $E(X^2)$ correctly as an integral in part (a). They then realised that, to evaluate the integral, they needed to differentiate by parts (twice). The standard of work was generally good with few errors. Those errors that did occur usually involved signs or missing brackets. Some students, who made integration errors, were awarded a mark for showing correct working for the variance, based on their knowledge of the exponential distribution. Part (b)(i) required students to 'derive' the distribution function. If it was merely quoted, then one mark only was awarded. Most students knew the appropriate technique, using logarithms, to solve part (b)(ii) correctly. A similar number had a reasonable idea of what was required in part (c) and so there was a good proportion of correct answers.

Q3.

The method marks in parts (a)(i) and (ii) were generally earned. Almost inevitably there were errors with signs, integration and use of logarithms, costing accuracy marks. Most candidates stated the correct probability in part (b)(i), but failed to recognise that it was needed in part (b)(ii)(A). The hope that 2 marks could be gained by writing down what was on the question paper was forlorn. Candidates needed to state that $P(T > t)$ was the probability that 0 strikes occur in the interval $(0, t)$ and hence that $P(T > t) = F(t) = 1 - e^{-t/\lambda}$ for $t \geq 0$. Most candidates knew that they needed to differentiate F with respect to t in order to obtain $f(t)$, but some then forgot to state the distribution of T , which was asked for in the question.

Q4.

The mathematical skills of most candidates were well able to cope with the pure mathematics involved in this question. In part (a), a lack of detail was evident in the answers. To gain both marks it was necessary to state $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ and $F(x) = 0$ otherwise; simply stating $F(x) = 1 - e^{-\lambda x}$ only obtained 1 mark.

Many candidates were able to find the two quartiles and hence the interquartile range in part (b). In part (c)(i), limits were required for the integration in order to score marks and, in fact, most candidates did use definite integration. As was the case in question 4, some lost the final mark for not showing a complete solution. Part (d)(i) was usually done successfully by using the printed answers but, in part (d)(ii), some candidates lost the final mark for an imprecise statement such as "it gets smaller".

Q5.

Part (a) produced considerable variation in the responses. Weaker candidates used incorrect limits, or omitted the limits completely, and sometimes candidates showed little knowledge of integration by parts. The more able candidates showed impressive knowledge of how to handle infinite integrals. As this is an applied unit, a somewhat informal approach, without the rigours of pure mathematics, to infinite integrals is acceptable. The specification requires candidates to know the form of the cumulative distribution function for the exponential distribution. Those who did were able to carry out the calculations in parts (b)(ii) and (iii) without any great difficulty. Again, the ability to deal with conditional probability was evident in part (b)(iii).

Q6.

For some candidates this was not a good first question but this certainly did not appear to affect their subsequent attempts at other questions. In part (a)(i), many candidates omitted the range $t < 0$ and/ or 0 for $t < 0$ and, in part (a)(ii), some candidates quoted 800^{-1} for the standard deviation. Nevertheless, this did not prevent them, and indeed almost all candidates, from scoring the 3 marks available in part (a)(iii). In part (b)(iv), only the most able candidates appeared to be aware that, for an exponential distribution, $P(T > t_2 | T > t_1) = P(T > t_2)$ but some other candidates were able to make significant progress, even if not with complete success, by the application of $P(A | B) = P(A \cap B) \div P(B)$. Whilst many candidates realised that part (b) required their answer to part (a)(ii), only the most able candidates also realised that a binomial distribution was then needed to evaluate $P(X = 2, 3, 4)$ or $1 - P(X = 0, 1)$.

Q7.

In part (a), better candidates stated “8” but the majority stated “ $\sqrt{8}$ ” or “0.125.” However part (b) was answered correctly by the vast majority of candidates. Sadly, the same was not the case in part (c). Candidates often treated the interval as one 30-minute period with $\lambda = 30$ (2×15) or 23 ($8 + 15$). Some candidates even tried the direct substitution of 5.45 and 6.15 into their exponential distribution function or attempted a normal linear combination. Even when two correct probabilities were calculated, they were all too often added rather than multiplied.

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